

BEDROCK ESG

Trinity Protocol — Human Reference Guide

v2.4 · Compiled June 16, 2026 · Confidential — Kato / SPEXTER Build Team

 **SPEXTER Umbrella — Locked**

● Indigo = Core Platform / Founder ● Blue = API / Admin ● Green = Live / Verified / Pass ● Cyan = SPEXTER / Field
● Purple = Blockchain ● Amber = Pending / Build Next ● Red = Rejected / Dropped

1. Platform Overview

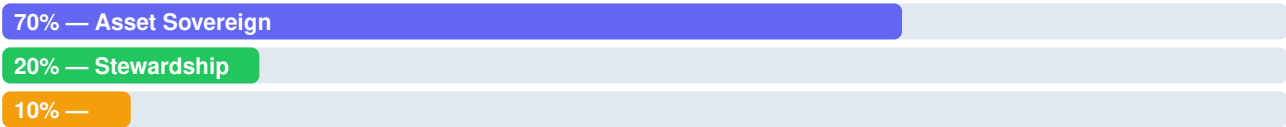
What Is Bedrock ESG?

An institutional-grade carbon verification platform — a digital notary that provides forensic-level proof of environmental impact. Three verticals: Asphalt Recycling, Biochar Carbon Removal, and Metal Recovery. Submissions are validated, scored, and anchored to three blockchains.

KEY METRICS



ZERO GREED SETTLEMENT LAW — HARD-CODED SPLIT



THREE CARBON VERTICALS



2. Technology Stack & Architecture

SIX ARCHITECTURE LAYERS

- 1 Presentation Layer** — Next.js App Router, 37+ pages, RBAC-gated middleware, custom JWT auth (no NextAuth)
- 2 API Layer** — 76 route handlers: forensic-ingest, telemetry, discovery, submission lifecycle, admin purge
- 3 Protocol Layer** — Aethexer Auth v1: Dual-Key + Timestamp ($\pm 300s$) + Nonce dedup. Sentinel Engine scoring.
- 4 Settlement Layer** — Universal Law (Zero Greed): 70/20/10 hard-coded split applied on all verified assets
- 5 Blockchain Layer** — Solana (notary memo) + Hedera HCS (submission records) + Polygon (ERC-1155 token minting)
- 6 Persistence Layer** — PostgreSQL: 45 Prisma models, 30 enums, 2,041 schema lines. Amazon S3 for file blobs.

KEY LIBRARIES & STANDARDS

Next.js App Router	Prisma ORM	AES-256-GCM	SHA-256 Magic Bytes	ICE v4.1
Ecoinvent v3.10.1	Eurobitume A1-A3	Haversine Formula	PLY/Splat Parser	IndexedDB + S3

3. Sentinel Engine — Three-Pillar Scoring

95% Document Integrity Magic byte validation, SHA-256 hash check, file count enforcement	75% AI Confidence LLM-based document analysis. Flags anomalies and inconsistencies.	75% Spatial Correlation Haversine GPS boundary check. 50km radius. Timestamp drift ≤15 min.
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VERDICT OUTCOMES

Verdict	Condition	Action
AUTO_APPROVED	All 3 pillars pass thresholds	Immediate settlement + blockchain anchor
SOVEREIGN_HOLD	Spatial fails, others pass	Queued for Founder manual review
REJECTED	GPS > 50km boundary	400 Bad Request — re-enter capture zone
PENDING_SOVEREIGN_REVIEW	Score below floor (0.72)	Manual admin queue

KNOWN FACILITY ZONES (FACILITY_ZONE_REGISTRY)

DC-SOVEREIGN Las Vegas, NV — 50km radius	DC-PROPERTY Houston, TX — 75km radius
DC-PALLETS Ontario, CA — 40km radius	DC-METALS Pittsburgh, PA — 60km radius



HUM Dynamic Listener

Monitors 12.6 MW energy market baseline. Triggers on-chain blockchain memo anchor when market shift exceeds 5%. Real-time energy market activity signal.

4. SPEXTER — Multi-Sensor Proof-of-Presence

What SPEXTER Actually Does

Spoof-resistant multi-sensor field verification. NOT 3D mesh scanning. Combines GPS coordinates, barometric altitude, geotagged image overlap, and device timestamps to prove physical presence at a registered facility. Phase 1 of forensic-ingest is LIVE at 588 lines with zero TypeScript errors.

WHAT'S REAL VS. WHAT'S BEING BUILT

✓ **REAL — Buildable Now**

🔧 **PIPELINE — Build Next**

Haversine GPS Geofencing

Standard spherical geometry. 50km hard boundary enforced.

/api/v1/forensic-ingest Phase 1

✓ BUILT — 588 lines. Haversine + Sentinel wired. Zero TS errors.

AES-256-GCM + IndexedDB

Bank-grade local encryption before transmission. Standard PWA pattern.

Confidence Scoring Pipeline

Baseline built. Needs full sensor snapshot weighting.

Geotagged Image Overlap

Sequential photos with timestamps + GPS metadata. Proves physical walkthrough.

\$EXITZ Solana Minting

Architecture spec complete. On-chain integration Phase 2.

Timestamp Drift Detection

Client vs server time delta ≤ 15 min. Catches temporal spoofing.

⊘ PERMANENTLY DROPPED

RF/Electromagnetic Signatures

iPhone does not expose raw RF data to apps. Removed from all documents.

SPEXTER INGEST PAYLOAD SCHEMA

```
operatorId: string
facilityId: string → resolves to FACILITY_ZONE_REGISTRY
clientTimestamp: ISO-8601 | Unix epoch ms
coordinates: { lat: float, lon: float }
sensorSnapshots: [{ sensorId, type, value, unit, capturedAt }]
→ Haversine check → Sentinel scoring → parserResultJson persist
```

5. Three-Chain Blockchain Architecture

🕒 Solana — Notary Anchor

Memo program writes forensic proof hash on-chain. Devnet active. Mainnet target for \$EXITZ minting. Phantom wallet integration planned.

🔗 Hedera HCS — Submission Records

Hedera Consensus Service stores submission record fingerprints. Dual attestation alongside Solana. Immutable timestamped consensus log.

🔗 Polygon — Token Minting

ERC-1155 token minting for verified carbon credits. \$EXITZ token ecosystem anchored here. Settlement triggers minting event.

\$EXITZ TOKEN FLOW

Field Capture → Sentinel ≥ 0.72 → Haversine Pass → 70/20/10 Split → \$EXITZ Mint

⚠️ **Minting Status:** Solana Devnet notary is active. \$EXITZ on-chain minting integration not yet coded — this is Phase 2 of the backend sprint.

6. Role-Based Access Control — 4-Tier Hierarchy

👑 **FOUNDER** — Full system access. Admin elevation via ADMIN_MASTER_EMAIL env var.

Level 100

🛡️ **ADMIN** — Discovery panel, purge gate, audit queue management. Gated by Founder elevation.

Level 75

🌐 **SOVEREIGN_AGENT** — Field operators. Can submit via Aethexer field key. SPEXTER data capture access.

Field

🏠 **CLIENT_OWNER** — View own submissions and settlements. No admin access.

Client

AUTH SYSTEMS

🔑 Web UI Auth

Custom JWT via auth-token cookie. Google SSO. No NextAuth dependency.

🤖 Field App Auth

Aethexer Protocol — x-aethexer-key + x-aethexer-nonce + timestamp $\pm 300s$ window.

ADMIN CONTROL PANEL — /ADMIN

Discovery
Ledger

Pending
Queue

Approve / Reject / Re-
Scan

PURGE Gate (typed
confirm)

12 Assets
Settled

7. Data Sovereignty — Option B (Clean Separation)

Formally codified in `lib/data-sovereignty.ts`. Five classification tiers. 26 registered data fields. Tier 3 reputation data is permanently BLACKOUT from all public streams.

Tier	Name	Public Visibility	Notes
Tier 1	SOVEREIGN_PRIVATE	✗ Never	Founder-only access
Tier 2	OPERATOR_PRIVATE	✗ Never	Field operator + admin only
Tier 3	REPUTATION_BLACKOUT	🚫 BLACKOUT	Excluded from all public flows permanently
Tier 4	CLIENT_VISIBLE	✓ Client-gated	Shown to authenticated CLIENT_OWNER
Tier 5	PUBLIC_STREAM	✓ Open	Anonymous public aggregator feeds

KEY LIBRARY FILES BUILT THIS SPRINT

File	Lines	What It Does
<code>lib/validator.ts</code>	—	38 HazardEventTypes. T1=0.50, T2=0.35, T3=0.15 weights. NOTARIZATION_FLOOR=0.72
<code>lib/data-sovereignty.ts</code>	—	5 DataClassification tiers, 26 fields, Tier 3 BLACKOUT enforced
<code>lib/spexter-aggregator.ts</code>	—	Hot-swap ready Tier 1 Esri ArcGIS mock adapter
<code>app/api/v1/forensic-ingest/route.ts</code>	588	Haversine enforcement, Sentinel wiring, SPEXTER field extraction

8. Build Status — What's Live vs. What's Next

Component	Status	Notes
Trinity Protocol Frontend	✓ LIVE	bedrockfad.com + bedrockesg.com
User Auth + RBAC (4 roles)	✓ LIVE	Custom JWT, Google SSO, middleware
Admin Control Panel /admin	✓ LIVE	Discovery, purge gate, ledger
Discovery Ledger (12 assets)	✓ LIVE	All 12 settled, 70/20/10 split applied
forensic-ingest Phase 1	✓ LIVE	588 lines, Haversine + Sentinel wired, zero TS errors
Sentinel Engine (3-pillar)	✓ LIVE	95/75/75 thresholds active
lib/validator.ts	✓ LIVE	38 HazardEventTypes, T1=0.50 hardcoded weights
lib/data-sovereignty.ts	✓ LIVE	5 tiers, 26 fields, Tier 3 BLACKOUT
lib/spexter-aggregator.ts	✓ LIVE	Esri mock adapter, hot-swap ready

MicroVault / Builders / Investors	 LIVE	Display-only / simulated data
bedrockfad.com domain	 LIVE	Deployed via Abacus, DNS verified
\$EXITZ Solana Minting	 PHASE 2	Architecture spec complete, not yet coded
On Point Dashboard UI	 OPEN	Not yet started
EXITZ Token Liquidity	 PENDING	Waiting on Kato decision
True Hardware Attestation	 FUTURE	Requires Apple DeviceCheck / Secure Enclave

9. Version History & Roadmap

- v1** **v1.0 — Scrapnet 2.0** — Asphalt RAP calculator. Original platform foundation.
- v2** **v2.0 — Bedrock ESG Rebrand** — Added Biochar, Metal Recovery, MicroVault, Blockchain anchoring.
- v2.2** **v2.2 — Settlement Rebrand** — Settlement label update. Precious Metals VAL formula added.
- v2.3** **v2.3 — Macro-Asset Discovery** — Discovery pipeline. Admin cockpit ticker. 12 macro-assets seeded.
- v2.4** **v2.4 — Trinity Protocol (CURRENT)** — RBAC, Audit Logs, Aethexer v1, Sentinel Engine, SPEXTER Phase 1 forensic-ingest, Haversine boundary enforcement, admin panel, bedrockfad.com live.
- v2.5** **v2.5 — SPEXTER Phase 2 + On Point (NEXT)** — \$EXITZ Solana minting integration. On Point Dashboard UI. Full SPEXTER sensor pipeline. Token liquidity strategy.

TRINITY STACK IDENTITY

Bedrock ESG — Carbon Verification
Layer

Aethexer — Field Protocol +
SPEXTER

Sovereign Skin™ + Gripsy on
Solana

Umbrella Brand: SPEXTER — Permanently Locked. 🏹